RISKLENS · DRILLING INTELLIGENCE

RiskLens for Drilling

Non-Productive Time is a real-time risk problem — not an operational afterthought. This deck is the response to your AI Implementation Engineer task.

DHANRAJ DESHMUKH · AI IMPLEMENTATION ENGINEER, OIL & GAS



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20–25%

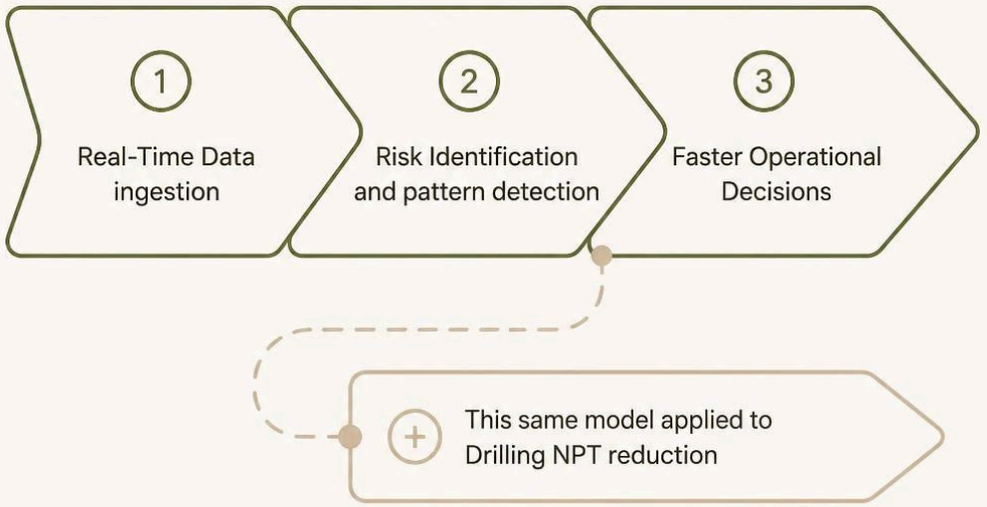
of all drilling time is lost to NPT — industry-wide, every year

Three failure modes account for the majority: **stuck pipe, well control incidents, and lost circulation.** Each one is predictable. None are being caught early enough.

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THE INSIDER MOVE

Same Model. New Vertical.



RiskLens Already Does This

Real-time data → risk identification → faster decisions. That is the core product model.

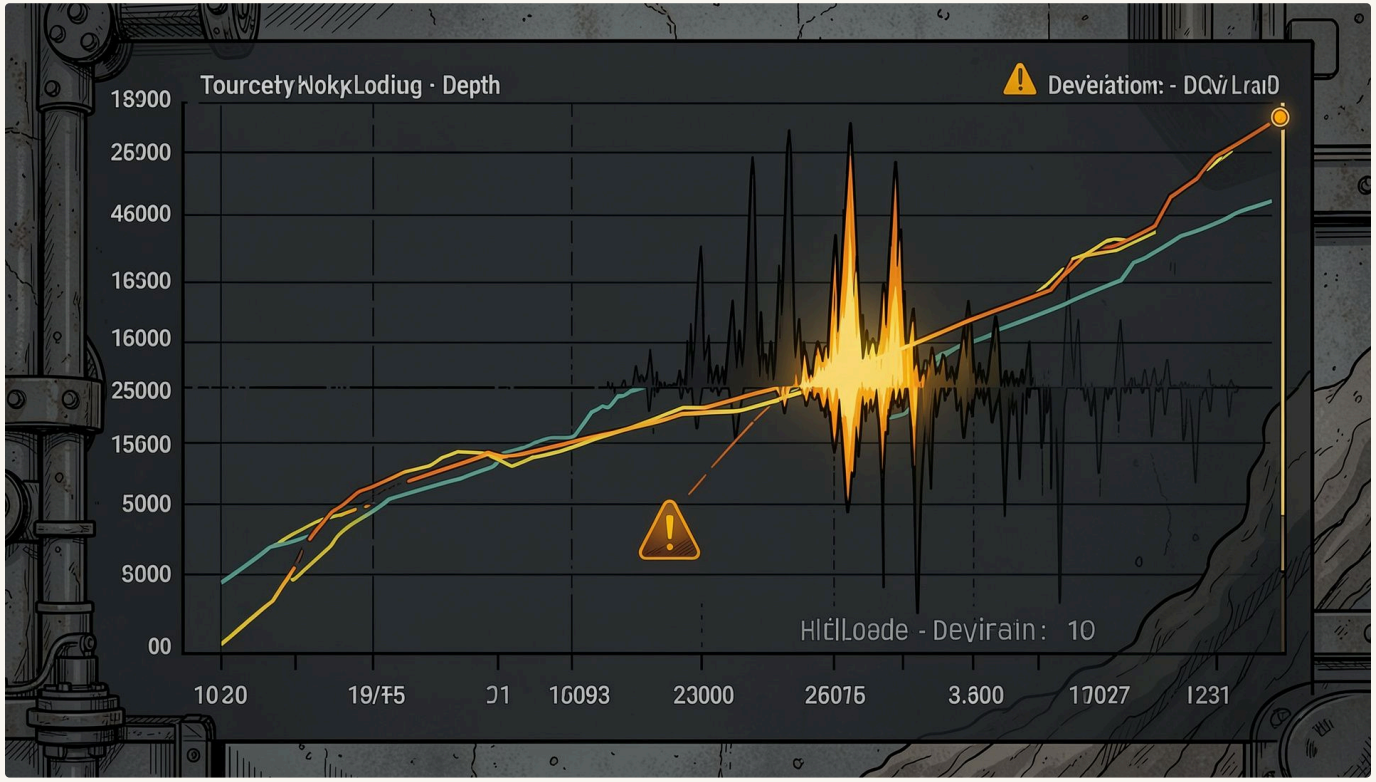
RiskLens for Drilling is not a new idea — it is the natural next vertical. The architecture is identical. The data sources are available. The use cases are proven.

This is an extension, not a pivot.

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MODULE 1

Stuck-Pipe Early Warning



Mechanism

ML monitors torque, hookload, and flow-in vs. flow-out in real time. Deviations from the baseline pattern trigger an alert **before** the pipe sticks — not after.

Proof

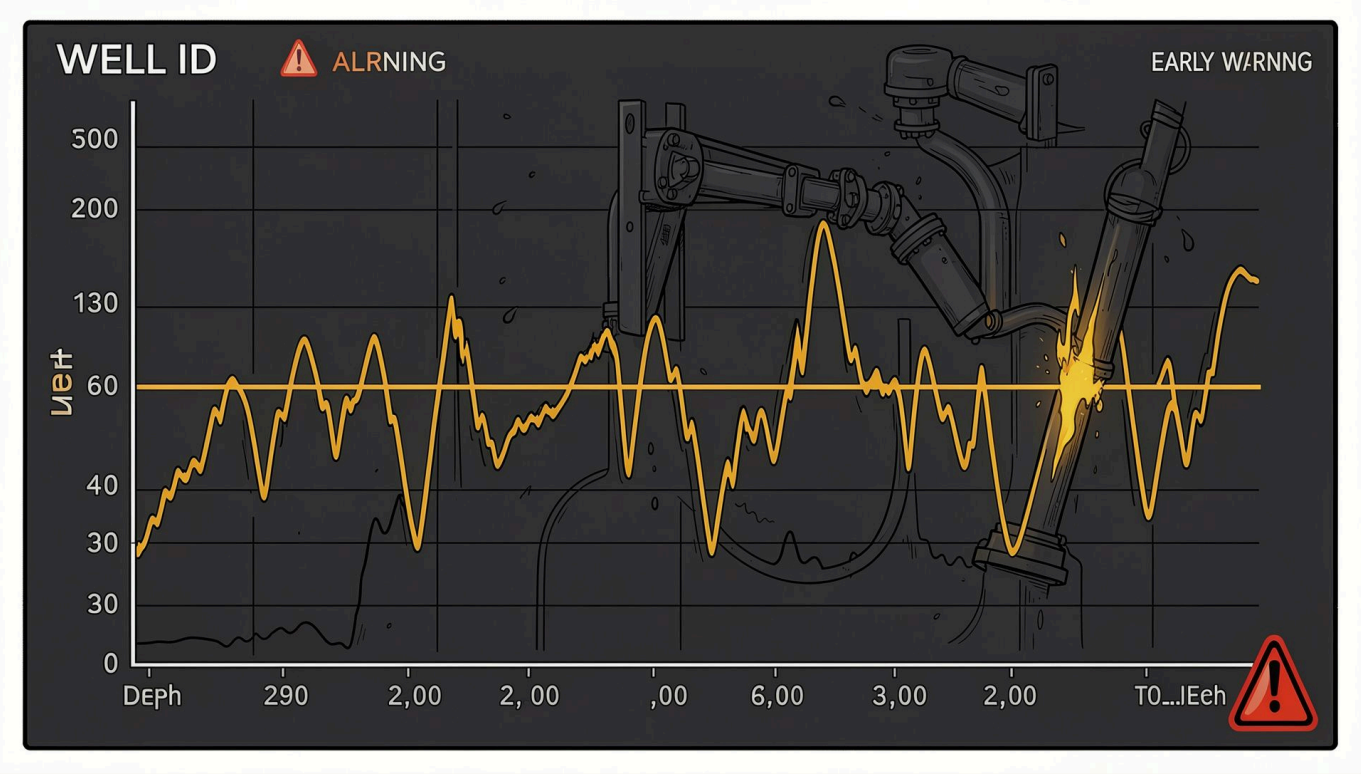
Saudi Aramco deployed an ML-based stuck-pipe early-warning system, reducing NPT events significantly on horizontal wells.

Aramco Journal of Technology, 2020

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MODULE 2

Well-Control Early Warning



Mechanism

LSTM-autoencoder models detect abnormal flow and pressure signatures — catching kick influx **9.2 minutes earlier** than conventional threshold alarms.

Proof

Pason Systems: ML-based alarm system detected **33 of 33 kicks** early across 80,000 rig-hours — zero misses.

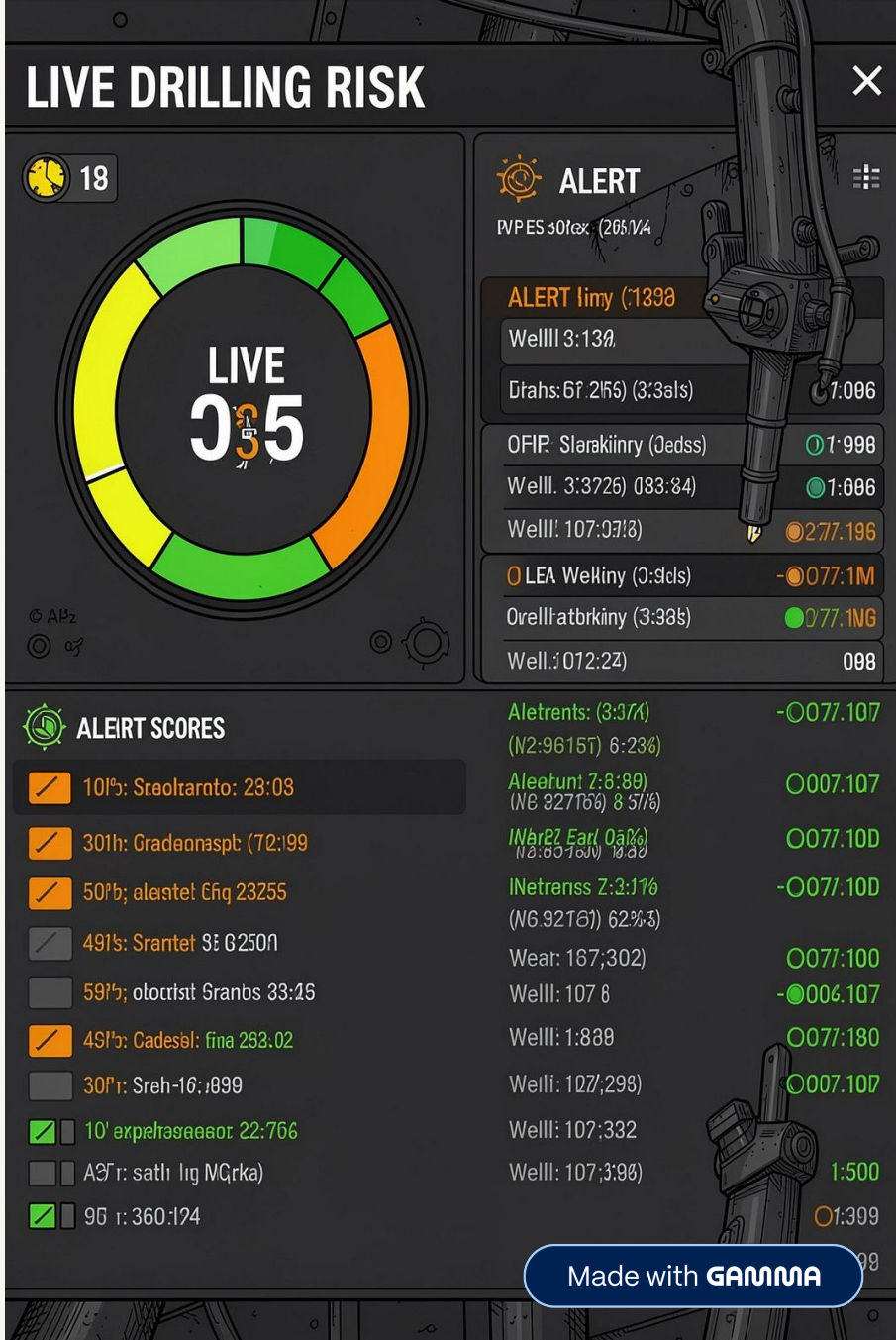
Petroleum Science, 2026 · Pason Systems Field Study

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Live Risk Console

One screen. All active wells. Risk scored, prioritized, and actionable — before the driller asks.

Concept interface — illustrative



EVIDENCE

Not a Bet — A Pattern Already Proven

60%

ROP Increase

SLB / Equinor autonomous drilling, Peregrino C platform

9.2min

Earlier Kick Detection

LSTM-autoencoder model, Petroleum Science, 2026

33/33

Kicks Caught Early

Pason Systems ML study, 80,000 rig-hours

weeks→hrs

NPT Report Review

Provectus GenAI analytics case study, 2025

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HONEST RISK

The Real Risk — and How We'd De-Risk It

The Risk

Data quality gaps and driller trust in a new system. No model works on bad inputs — and no alert works if the crew ignores it.

The De-Risk Move

Deploy in **advisory-only mode** on 1–2 rigs for 90 days. No automated actions. Pure visibility. Let the data earn trust before the system takes control.

- Baseline NPT hours tracked from Day 1
- Driller feedback loop built into alert tuning
- Escalation to automated response only after validation

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The Ask

A 90-day advisory-mode pilot on one rig.

Measured against baseline NPT hours. Zero automated actions. Pure risk visibility — and a clear before/after story at the end.

Dhanraj Deshmukh · AI Implementation Engineer, Oil & Gas

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